Exponent study: does the subgroup-base algebraic route have an

exponent lever? (2026-11-10) **Class.** An engine measurement in counted units: F_p multiplications of the F4 row reduction, `F4Report.field_ops`, from `curve25519_family f4sub`. It measures the cost of one decomposition system against the base size. It is not an end-to-end solve, not an ecbench session, and not a leaderboard or `benchmarks/` row. No pcc claim was registered, so every measured number below is an unverified assertion. Rho remains ahead on every track [pcc:unverified].

Question (owner preregistration, 2026-11-08). The question concerns Algorithm 1 of the algebraic ECDLP attack on prime curves (Petit, Kosters, Messeng, PKC 2016) and the cost model of its section 3.2. The total cost is about P(p,m) + (m! p / (deg L)^(m-1)) T(E,m,L) + (deg L)^omega, with deg L = |V|. With omega = 2 and |V| = p^a, m = 3 balances at a = 1/4 and m = 4 at a = 1/5. An exponent lever exists at m = 4 only if T(E,4,L) grows more slowly than |V|^(1/2). At m = 3 it exists only if T stays flat in |V|. This study fits the exponent s of T against |V|. It uses the symmetrised system over multiplicative-subgroup bases of the Curve25519-family miniatures.

Answer. No lever was found at m = 2 [pcc:unverified]. At m = 3 and m = 4 the frozen resource limits censored the series before the preregistered estimator could apply, so the verdict there is "undetermined" [pcc:unverified]. The limits were 3.5 GB of virtual memory per process (`ulimit -v`) and 1,200 s of wall time per target, on a shared 4-core host. This is a resource limit, not evidence that no lever exists at larger |V|.

- **m = 2.** The preregistered fit gives s = 2.77 [2.31, 3.23] at k = 20, 2.81 [1.81, 3.80] at k = 22 and 2.57 [0.91, 4.24] at k = 24 [pcc:unverified].
- **m = 3 and m = 4: the preregistered estimator could not be applied.** Before the first censored cell, the complete cells with d >= 9 span 3.0x in |V| at k = 20 and 3.7x at k = 22, below the required 4x. At m = 4 only one such cell is complete. The preregistered verdict is therefore "undetermined" for both [pcc:unverified].
- **Other readings.** The m = 3 sensitivity fits add the boundary cells d < 9 (d = 3 and 7 at k = 20, d = 3 and 5 at k = 22). They give 6.44 [3.49, 9.39] at k = 20 and 5.76 [3.77, 7.75] at k = 22, and they are the only m = 3 readings whose lower bounds lie above 0 [pcc:unverified]. The d >= 9 fits, computed despite the span gate, give 5.56 [−8.75, 19.88] at k = 20 (4 cells) and 4.51 [−1.27, 10.29] at k = 22 (6 cells), so the d >= 9 cells alone do not exclude s < 0 at 95% at m = 3 [pcc:unverified]. At m = 4 the only fit, over the three cells d = 3, 7 and 9, gives 9.86 [−59.9, 79.6] [pcc:unverified]. Every point estimate lies far above the m = 3 threshold (s < 0) and the m = 4 threshold (s < 0.5), and no upper bound lies below them, so no reading meets the lever rule [pcc:unverified]. From |V| = 3 to |V| = 9, the m = 4 F4 cost T rises 3.4e5-fold; T excludes the per-target construction of the symmetrised system (section 8) [pcc:unverified].

Protocol and companion files. `PROTOCOL.md` was frozen on 2026-11-08, before any measurement. Its sha256 is in `data/frozen.sha256`. On 2026-11-10 the hashes of the protocol, `scripts/run_grid.py`, `scripts/analyse.py` and the engine binary were recomputed, and all four equal the frozen values [pcc:unverified]. `PRIOR_STUDIES.md` is an addendum written after the grid had run. The report is `REPORT.pdf` and the figure is `exponent_fit.svg`. The independent validation is `validator-report.json` its errata are applied in the section "Validation" at the end.

Post-data change to a non-frozen script. On 2026-11-10, after all data were taken, the non-frozen `scripts/tables.py` was edited in commit `8c544a1829`. The edit changes only the status label of a censored cell in `data/tables.md`, which now names the cause ("1200 s cap" or "OOM at 3.5 GB") instead of always "OOM". No number, fit or verdict depends on it, and the frozen `scripts/analyse.py` was not touched.

Source, toolchain and binary. The engine binary was built from the tree at commit `8d0388b10a` (the protocol freeze, 2026-11-08T23:50:48Z), before the merge `51b1ba83bc`. A `cargo build` at a later HEAD gives a different binary. The toolchain on the host is `rustc 1.97.0 (2d8144b78 2026-07-07)` (`cargo 1.97.0`), read on 2026-11-10; the build itself did not record its toolchain. The frozen binary, sha256 `dc2c30a4242d80281622b2cb42cbaf9cb457d041ffbc9a4d35794bb893bf3faf`, was copied out of the shared build directory to `/home/user/bin-archive/curve25519_family-dc2c30a4` on 2026-11-10, and the copy has the same sha256. It is not committed to Git and not yet in the artifact store.

Resume after a container restart (2026-11-10). A container restart killed the grid runner while it was measuring `k20_m3_sym_d27_g0`. The runner writes a cell file only when all six targets of the cell are done, so no partial data from that cell exists or was used. All other cell files were complete or censored before the restart, and none was redone. The interrupted cell was rerun from scratch with the frozen binary. The binary was a copy of `/home/user/ctarget-f1/release/curve25519_family`, with sha256 `dc2c30a4242d80281622b2cb42cbaf9cb457d041ffbc9a4d35794bb893bf3faf` (equal to the frozen hash). The binary is newer than `subgroup_semaev.rs`, `pkm_semaev.rs` and `src/bin/curve25519_family.rs`, so it was not rebuilt. The rerun completed: 6 of 6 targets, each under the 1200 s cap. Under the chain rule the runner then ran d = 57, whose first target hit the 1200 s cap, so that cell is censored. The k = 22 m = 3 cell at d = 55 had ended in an out-of-memory exit before the restart (`data/run.log`) [pcc:unverified].

Host, load and wall time. The runner logged no host name, core count or concurrent load. The censoring rule is wall clock (monotonic) on a shared 4-core host. Per the session log, the d = 27 rerun and the censored d = 57 target ran while another agent built `ecbench` on the same host. The d = 57 target ran at about 6.9e7 field ops/s against about 2.9e8 for the d = 27 rerun, so its censoring may partly reflect host load [pcc:unverified]. There are also unexplained wall-clock gaps: `k22_m3_sym_d55` was

written about 26.5 h after d41, and the original d = 27 attempt ran about 27 h without writing a cell file, against about 1.5 h for the rerun [pcc:unverified]. These fit a VM pause that the monotonic budget did not see, but that cannot be checked. Wall time is not a claimed quantity of this study; only field_ops is.

1. System and engine

- **Module.** src/cryptanalysis/ic_framework/subgroup_semaev.rs, commands curve25519_family f4sub and subcensus. It is a separate module rather than a new Membership variant of pkm_semaev.rs: the tower engine's bit-mask monomials hold exponents 0 or 1 only, so they cannot hold the odd-degree chain rules y^3 = y'. No path of pkm_semaev or f4_fp_tower changed.
- **Base.** V = lo + g mu_d, d | p - 1 odd, d = e_0 e_1 ... e_{L-1} (primes ascending, with multiplicity), lo = A/3 (the Montgomery frame), g = w^gidx with w the smallest primitive root.
- **Per-summand constraint.** This is the power-map chain of the paper: x_i = lo + g y_{i,0}, y_{i,j}^{e_j} = y_{i,j+1} for j < L - 1, and the closing relation y_{i,L-1}^{e_{L-1}} = 1. The paper's closing map is 1 - x^{p_n'}; this is the same relation.
- **Target and form.** The target X = x(aG) is a constant, with a uniform in [1, r - 1]. The form sym (primary) adds e_k - sigma_k(x_1..x_m) for k = 1..m and the Semaev polynomial S_{m+1}(x_1..x_m, X), written in e_1..e_m by interpolation. The interpolation is checked on 12 extra points. The form plain substitutes sigma_k(x) for e_k (ablation).
- **Engine.** f4_fp::f4, grevlex, max degree 400. There is no guessing (G = 0): T is the cost of one complete system.
- **Unit tests.** The module has six tests. They cover: the summation polynomials vanish on sums (m = 2, 3, 4); S equals S (m = 2, 3, 4); roots of unity and the chain factorisation (roots_of_unity_and_factors); the systems on p = 211 (m = 2 with d = 7 and 15, m = 3 with d = 7) are consistent if and only if S_{m+1} has a root in V^m, and a sum of base points always gives a consistent system; the bit-interval base matches brute force. The tests do not assert that a system is consistent exactly when the target is a sum of m base points; for m >= 3 twist points can also give roots. No m = 4 consistency test exists. The producer ran them from the release test binary built at the module's commit (6 passed, 0 failed) [pcc:unverified]. The test log was not committed, and the tests were not rerun for this revision, because building was not allowed on the host.

2. Grid and census

The grid is as frozen in section 4 of the protocol:

- 6 targets per cell, with seed 20261108 + 1000 m + d;
- a cap of 1200 s per target and 3.5 GB of memory per process;
- each (k, m, form) chain runs d ascending and stops at its first censored cell. Larger d are not attempted.

Every chain in the protocol has now ended, at its last d or at a censored cell. The census (data/census.json, from scripts/census.py) is an exact enumeration [pcc:unverified]. Its columns are defined as follows:

- **columns:** the number of abscissae of V on the curve;
- **base points:** the number of signed base points;
- **covered:** the number of distinct points of the order-r subgroup that are sums of exactly m base points.

Almost every target lies outside the covered set, so almost every system is inconsistent, as in the real attack: 8 of the 414 measured systems are consistent [pcc:unverified]. At m = 3 the covered set is empty for d = 3 and 9 at k = 20 and for d = 3 at k = 22 [pcc:unverified]. T is therefore mostly the cost of proving that a system has no solution [pcc:unverified].

The four curves are measured from data/census.json:

k	p	r
18	262,133	32,887
20	1,048,573	131,203
22	4,194,301	524,731
24	16,777,213	2,097,349

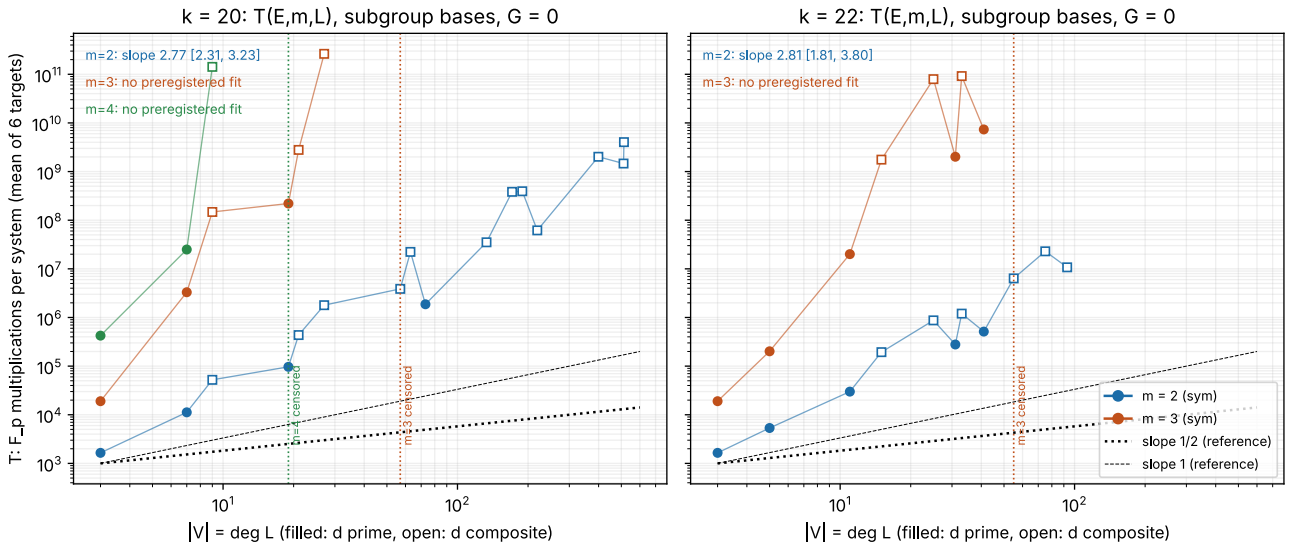
The full per-cell table is in data/tables.md, with the census, T, the solving degree and the largest matrix width. The primary series are below. T is the mean field_ops per system and L is the number of chain levels. T is in F4 field_ops only (section 8 lists the unpriced work). All values in the table are measured, from cells/*.json and data/census.json [pcc:unverified].

k	m	d = |V|	L	columns	covered (m)	T	solving degree	status
20	2	3	1	2	2	1,648	4	complete
20	2	7	1	2	2	11,225	8	complete
20	2	9	2	4	8	52,244	5	complete
20	2	19	1	12	40	96,745	20	complete

20	2	21	2	8	20	434,893	9	complete
20	2	27	3	16	68	1,789,928	5	complete
20	2	57	2	34	314	3,876,409	21	complete
20	2	63	3	28	200	22,317,287	9	complete
20	2	73	1	36	328	1,869,761	74	complete
20	2	133	2	76	1,452	35,142,880	22	complete
20	2	171	3	108	3,024	383,236,931	21	complete
20	2	189	4	96	2,288	393,654,078	10	complete
20	2	219	2	120	3,552	61,803,451	75	complete
20	2	399	3	216	11,254	2,016,013,118	22	complete
20	2	511	2	274	17,640	1,463,328,780	76	complete
20	2	513	4	278	18,564	4,027,107,095	22	complete
20	3	3	1	2	0	19,010	4	complete
20	3	7	1	2	4	3,303,793	8	complete
20	3	9	2	4	0	147,635,021	7	complete
20	3	19	1	12	144	220,436,708	20	complete
20	3	21	2	8	56	2,787,948,088	11	complete
20	3	27	3	16	344	263,459,825,507	8	complete (rerun after restart)
20	3	57	2	34	3,124	-	-	censored (1200 s cap, first target)
20	4	3	1	2	4	423,322	8	complete
20	4	7	1	2	4	24,975,907	8	complete
20	4	9	2	4	24	142,058,000,763	9	complete
20	4	19	1	12	504	-	-	censored (out of memory)
22	3	3	1	3	0	19,010	4	complete
22	3	5	1	3	4	201,745	6	complete
22	3	11	1	7	32	20,063,347	12	complete
22	3	15	2	9	52	1,759,715,051	9	complete
22	3	25	2	19	576	79,458,456,936	10	complete
22	3	31	1	11	84	2,016,009,937	32	complete
22	3	33	2	17	420	91,887,410,934	15	complete
22	3	41	1	23	1,020	7,342,801,721	42	complete
22	3	55	2	29	2,076	-	-	censored (out of memory)

The censored k = 20, m = 3, d = 57 target had spent 8.3e10 field operations when it was stopped at degree 23 with an 777,369 x 798,826 matrix (`cells/k20_m3_sym_d57_g0.json`). That partial count is not a T value and is not fitted [pcc:unverified]. The m = 2 series at k = 22, 24 and 18, the plain-form ablations and the coset cells are in `data/tables.md`.

Figure. The figure plots T against |V| on log-log axes, with k = 20 on the left and k = 22 on the right (`exponent_fit.svg`, from `scripts/figure.py`). Filled markers are prime d and open markers are composite d. The dotted vertical lines mark censored cells, and two reference slopes (1/2 and 1) are drawn.



3. Fitted exponents

The preregistered estimator is the OLS fit of ln T on ln |V| over the complete cells with d >= 9, with a two-sided 95% t-interval. It requires at least 4 cells spanning at least 4x in |V|. Its results are in `data/analysis.json`, written by the frozen `scripts/analyse.py`. The table's fits come from the cells through that script, except the d >= 9 fits in the last column for the gated series, which the validator computed despite the span gate (`validator-report.json`, C11 and E1). For m = 4 that column holds the validator's fit over all three complete cells.

series	cells in the fit	s	95% interval	sensitivity fit (all d)	d >= 9 fit despite the gate
k = 20, m = 2, sym	14	2.77	[2.31, 3.23]	2.80 [2.47, 3.12] (16 cells)	(primary)
k = 22, m = 2, sym	9	2.81	[1.81, 3.80]	2.72 [2.23, 3.21] (11)	(primary)
k = 24, m = 2, sym	4	2.57	[0.91, 4.24]	2.43 [1.89, 2.97] (5)	(primary)
k = 20, m = 2, plain	6	3.20	[1.67, 4.72]	3.31 [2.65, 3.97] (8)	(primary)
k = 18, m = 2, sym	3 cells in all	no fit	-	local slopes 2.19, 3.91	-
k = 20, m = 3, sym	4 (d = 9 to 27, span 3.0x)	no fit (span < 4x)	-	6.44 [3.49, 9.39] (6)	5.56 [−8.75, 19.88] (4)
k = 22, m = 3, sym	6 (d = 11 to 41, span 3.7x)	no fit (span < 4x)	-	5.76 [3.77, 7.75] (8)	4.51 [−1.27, 10.29] (6)
k = 20, m = 3, plain	3 before censoring	no fit	-	6.48 [2.45, 10.52] (5)	-
k = 20, m = 4, sym	1 (d = 9)	no fit	-	local slopes 4.81 (3 to 7), 34.4 (7 to 9)	all three cells: 9.86 [−59.9, 79.6]

The fits in this table are derived from measured cells [pcc:unverified]. At m = 3 only the sensitivity fits, which include the boundary cells d < 9, have lower bounds above 0; the d >= 9 fits do not exclude s < 0 at 95% [pcc:unverified]. The local slopes of the k = 20, m = 3 series are 6.09, 15.1, 0.54, 25.4 and 18.1, in order of d (`data/analysis.json`) [pcc:unverified].

Exploratory, not preregistered: the chain length matters. T depends strongly on how d factors, not only on |V|. At equal |V|, a composite d (more chain levels) costs more than a prime d. For example, at k = 20 and m = 2, T is 1.87e6 at d = 73 (L = 1) and 2.23e7 at d = 63 (L = 3) [pcc:unverified]. This explains some negative local slopes in the preregistered series. Fits restricted to one chain length are tighter. They come from `scripts/tables.py` and are stored in `data/exploratory_fits.json`.

- m = 2, L = 1 gives 2.20 [2.15, 2.25] at k = 20 (d = 3, 7, 19, 73), 2.19 [2.14, 2.23] at k = 22 (5 cells) and 2.21 [2.13, 2.28] at k = 24 (3 cells) [pcc:unverified].
- m = 2, L = 2 gives 2.43 [2.10, 2.77] at k = 20 (6 cells) [pcc:unverified].
- m = 3, L = 1 gives 4.95 [4.51, 5.40] at k = 22 (d = 3, 5, 11, 31, 41) and 5.04 [−1.79, 11.88] at k = 20 (d = 3, 7, 19) [pcc:unverified].

Exploratory, computed by the validator: regression on |V| and L together. The validator fitted ln T on ln d and L over all complete cells of each series (`validator-report.json`, C10). The slopes in ln d are 2.40 [2.21, 2.59] at k = 20, m = 2; 2.22 [2.03, 2.41] at k = 22, m = 2; 2.19 [2.02, 2.36] at k = 24, m = 2; 4.77 [3.35, 6.20] at k = 20, m = 3; and 4.98 [4.40, 5.55] at k = 22, m = 3 [pcc:unverified]. The L coefficient excludes 0 in every series (1.01, 1.29, 1.18, 2.68 and 3.84) [pcc:unverified]. Adding L lowers every exponent, but every slope stays far above the thresholds (s < 0 at m = 3, s < 0.5 at m = 4) [pcc:unverified]. The preregistered single-regressor estimator stays primary; this fit is exploratory.

The solving degree is d + 1 for prime d in the sym form at m = 2 and m = 3 [pcc:unverified]. It is not d + 1 in 7 prime-d cells: k = 20, m = 4, sym, d = 3 gives 8; the plain form gives 5, 9 and 21 at m = 2 (d = 3, 7, 19) and 12, 15 and 27 at m = 3 (d = 3, 7, 19) [pcc:unverified]. Where it holds, the cost is that of one dense equation of degree about d in each chain variable [pcc:unverified].

Comparison with the tower base (L10). L10 measured m = 2 on the isogeny factor base of the tower engine, with guessing G = t - 1. There T grows as about |V|^1.5: mpt 17,592, 52,816 and 146,464 at t = 4, 5, 6, in tower-engine mul-adds. The G = 0 comparison below did not use L10's base: it ran the bit-interval base, on the same curve (k = 20), in both engines. The tower engine's `f4cost` gives 87,013, 4,046,086 and 127,284,443 mul-adds per target at t = 3, 4, 5 (`data/calib_tower_bits_t*.json`) [pcc:unverified]. These are local slopes of 5.5 and 5.0 over |V| = 2^t = 8 to 32, in tower-engine mul-adds (derived) [pcc:unverified]. This engine's `f4sub` gives 294,981, 10,458,148 and 178,989,665 field_ops on the same base (`data/calib_sub_bits_t*.json`), local slopes of 5.15 and 4.10 in field_ops over the same |V| range (derived) [pcc:unverified]. The subgroup base's s = 2.77 is a fit over |V| = 9 to 513, so the slopes cover different ranges. At G = 0 and equal |V| the subgroup base is cheaper than the bit-interval base, both in field_ops: 1.79e8 at |V| = 32 (bits) against 1.79e6 at d = 27 [pcc:unverified]. L10's slope with guessing on the isogeny base, about 1.5, is below this base's G = 0 slope of 2.77 (2.20 at L = 1), so L10's guessing strategy grows more slowly than the subgroup base without guessing [pcc:unverified]. All of these slopes are far above the 1/2 that a lever at m = 4 needs [pcc:unverified].

4. Balanced totals under the paper's model (derived)

The relation phase costs p^(1 - (m-1)a) T(p^a) = p^(1 - (m-1)a + s a), and linear algebra costs p^(2a). The protocol's balance point is a = 1/(m + 1 - s), with total exponent 2/(m + 1 - s). When s >= m - 1, the relation-phase exponent 1 + (s - (m - 1)) a does not decrease as a grows. No base size then gives a total below p^1, and the balance either lands above 1 or has no positive solution.

m	s used	2/(m + 1 - s)	best total exponent under the model
2	2.7732 (k = 20 fit)	2/0.2268 = 8.82	1 (no base size helps; s >= m - 1 = 1)
2	2.8064 / 2.5745 (k = 22 / 24)	10.33 / 4.70	1
3	6.44 / 5.76 (sensitivity fits)	no positive solution (m + 1 - s < 0)	1 (s >= m - 1 = 2)
3	4.95 (k = 22, L = 1)	no positive solution	1
4	4.812 (local slope, 3 to 7)	2/0.188 = 10.64	1 (s >= m - 1 = 3)
4	34.4 (local slope, 7 to 9)	no positive solution	1

All values in this table are derived from fixed small p and are extrapolations, not measurements [pcc:unverified]. The balanced-total exponents use the unrounded slopes, because the value of 2/(m + 1 - s) is sensitive to rounding near its pole. A total below p^(1/2) at m = 4 would need s < 0.5 at the paper's a = 1/5, or s < 1 at the re-balanced a. A total below p^1 would need s < 3 [pcc:unverified].

5. Verdicts on the preregistered thresholds and predictions

item	preregistered rule	outcome
lever at m = 4	s < 0.5 with upper 95% bound < 0.5	**undetermined** by the estimator: one complete cell with d >= 9, and d = 19 censored. Not met on any reading (local slopes 4.81, 34.4)
lever at m = 3	s < 0 with upper bound < 0	**undetermined** by the estimator: span 3.0x at k = 20 and 3.7x at k = 22, both below 4x. Not met on any reading: sensitivity fits 6.44 [3.49, 9.39] and 5.76 [3.77, 7.75]; d >= 9 fits 5.56 [−8.75, 19.88] and 4.51 [−1.27, 10.29], whose lower bounds do not exclude s < 0
looser derived m = 4 break-even	s < 1	not met on any reading
prediction m = 2: s = 1.5 (range 1.0 to 2.3)		**failed**: 2.77 [2.31, 3.23] at k = 20, outside the range
prediction m = 3: s = 3.5 (range 2.5 to 5)		not evaluable by the estimator. The readings 6.44 and 5.76 are above the range; the L = 1 fits, 4.95 and 5.04, are inside it
prediction m = 4: s >= 4.5 if measurable (most likely capped beyond d = 9 or 21)		**undetermined** for s (not measurable: no fit exists). Only the censoring expectation held: out of memory at d = 19
plain vs sym: |s_plain - s_sym| < 0.5		**held**: 3.20 vs 2.77, a difference of 0.43

coset dependence: T within 1.5x across gidx		**held**: at d = 27, T is 1,789,928 / 1,789,919 / 1,789,928 for gidx = 0, 1, 2; at d = 63 it is 22,317,287 / 22,317,373 / 22,317,372
calibration: field_ops / f4_muladds within 5x		**held**: 3.39, 2.58 and 1.41 at t = 3, 4, 5
calibration: that ratio constant to 30%		**failed**: it falls from 3.39 to 1.41
owner's expectation: no lever		consistent: no reading comes near a threshold

The outcomes in this table come from measured cells and the fits derived from them [pcc:unverified]. Failed predictions stay failed.

Protocol eligibility ambiguity. The protocol's clause "with at least 4 complete cells whose |V| spans at least 4x" can also be read as gating on all complete cells, not only on those with d >= 9. The frozen `scripts/analyse.py` gates on the d >= 9 cells, and this study follows it, so "undetermined" is the preregistered reading. Under the other reading the d >= 9 fits (5.56 and 4.51 above) would be computed, and the m = 3 lever would read "not met" instead of "undetermined" [pcc:unverified]. No lever is found under either reading [pcc:unverified].

6. What a lever would need

Under the paper's model, a lever at m = 4 needs a decomposition system whose cost grows more slowly than |V|^(1/2). At m = 3 it needs one whose cost does not grow at all. Measured here, the cost grows as |V|^2 or more steeply at m = 2 on every series [pcc:unverified].

Trivial enumeration of V^(m-1), with a root test for the last summand, costs |V|^(m-1). At m = 4 that is already |V|^3, six times the exponent a lever needs. A lever would therefore need a solver that neither enumerates nor reduces over the |V|^m quotient of the chain ideal. This study found no such solver [pcc:unverified].

The m = 2 slope of the prime-d cells, about 2.2, is above even the enumeration exponent 1. This suggests that the F4 route over the chain ideal costs more than enumeration on this base (derived) [pcc:unverified].

7. Prior exponent studies

`PRIOR_STUDIES.md` lists five earlier studies.

- `ic_exponent_20260926` fits the ratio to batch rho on binary Koblitz curves.
- `ic_m4_exponent_audit_20260928` measures c = 0.985 bits per unit n for the m = 4 decomposition over an F_2-subspace base of binary fields.
- `ic_exponent_top_20260930` was withdrawn before anything ran.
- `ic_exponent_falsifiers_20261013` studies the exponent sum of base abscissae, not a cost.
- `ecdlp_exponent_assessment_20261018` has PDFs that could not be retrieved on this machine.

None of the readable four measures T(E, m, L) against the size of a prime-field multiplicative-subgroup base, so none gives a number to compare directly with s here. The closest is the m = 4 audit, on a binary field with an F_2-subspace base. It measured c = 0.985 bits per unit n [0.983, 1.018] for the m = 4 decomposition and recorded that route as "closed for this engine at these sizes" (as summarised in `PRIOR_STUDIES.md`). Its verdict points the same way as this study's [pcc:unverified]. The m = 2 tower-base values of L10 are compared in section 3.

8. Distance to rho

This study does not produce an end-to-end cost. As an illustration only, the modelled cost is E = columns · r · T / covered, in F4 `field_ops` (F_p multiplications). The rho reference 1.5 sqrt(r) counts group operations, not field_ops. No conversion factor between the two units was recorded or measured, so each ratio below is "field_ops per rho group operation", not a ratio of equal units. At k = 20 and m = 2, the ratio of E to 1.5 sqrt(r) ranges from 3.98e5 field_ops per rho group operation (d = 3) to 1.46e10 (d = 513) (derived) [pcc:unverified]. Native operations are unpriced, so the w band of AGENTS.md section 5a is not applied. The ratios are not comparable with L10's mul-add figures beyond the calibration ratio of 1.4 to 3.4. Rho remains ahead [pcc:unverified].

Unpriced work. T is the F4 row reduction only, as the protocol defines it. The following work is not counted in T or in E:

- **Per-target construction of the symmetrised system.** `build_system` calls `symmetrise` for every target, interpolating S at that target X by Gauss-Jordan elimination on an (n + 12) x (n + 1) matrix, with n = 6, 35 and 495 monomials at m = 2, 3 and 4. The validator's count is about 4.6e2, 3.1e4 and 6.2e7 F_p multiplications per target [pcc:unverified]. At m = 4 this exceeds T at d = 3 (4.2e5) and d = 7 (2.5e7); at m = 3 it exceeds T at d = 3 (1.9e4) [pcc:unverified]. It is constant in d and could be amortised to once per curve by interpolating in (e, X).
- **Rho's native work and the conversion of group operations to field_ops** (above).
- **Census and system specialisation** outside F4 (enumeration of V and the covered set), which is an analysis aid, not part of T.

9. Not done and open issues

- **m = 3 and m = 4 exponents.** Neither could be fitted under the frozen rule. A wider span at m = 3 needs more than the frozen 3.5 GB and 1200 s per target. At k = 22, d = 55 exited out of memory. At k = 20, d = 57 hit the 1200 s wall-clock cap at degree 23, with a matrix of about 8e5 columns, on a host shared with another agent's build (see "Host, load and wall time"). A sparser linear-algebra back end would also help. This is a resource limit, not a result.
- **Process items.** No `bench.record/v1` record exists, and the grid ran locally rather than on the EKS route. These are for the coordinator.
- **m = 4 stays open.** It has three complete cells (d = 3, 7, 9) and is censored at d = 19.
- **Chain length.** The protocol did not anticipate that prime d and composite d behave differently. A follow-up should fix L, or regress on (ln |V|, L), in its preregistration, and should name the lower envelope over factorisations (prime d is cheapest at equal |V|) as the quantity relevant to a lever.
- **Inconsistent systems.** Most targets give an inconsistent system, so T measures refutation cost. Consistent and inconsistent systems were not split, because the protocol does not ask for it.
- **Unretrieved prior study.** The `ecdlp_exponent_assessment_20261018` PDFs live in the artifact store and were not retrievable on this machine.
- **Unavailable tools.** research-kb, ecdb and Conductor were unavailable in this session: there was no MCP server and no token, and Conductor is not installed. The sources checked were the repository files named in `PROTOCOL.md` and `PRIOR_STUDIES.md`.
- **No pcc claims.** None were registered, so every result is an unverified assertion until checked.

Files

file	what
`PROTOCOL.md`, `data/frozen.sha256`	the frozen protocol and the hashes of the protocol, the scripts and the binary
`PRIOR_STUDIES.md`	the addendum on the five earlier exponent studies
`validator-report.json`	the independent validation (errata E1 to E13), committed unchanged
`scripts/run_grid.py`	the grid runner (resumable per cell)
`scripts/analyse.py`	the preregistered estimator; writes `data/analysis.json`
`scripts/census.py`	the exact census; writes `data/census.json`
`scripts/tables.py`	`data/tables.md` and the exploratory fits at fixed L (`data/exploratory_fits.json`)
`scripts/figure.py`	`exponent_fit.svg` and `.png`
`cells/*.json`	one file per cell, with per-target field_ops, solving degree, rows and columns
`data/calib_*.json`	the calibration cells (bit-interval base, both engines)
`data/run.log`	the runner's log, with the resume note of 2026-11-10
`REPORT.typ`, `REPORT.pdf`	the report (pandoc to Typst, compiled with the typst Python module)

Reproduce: use the archived frozen binary (sha256 `dc2c30a4…3faf`, see "Source, toolchain and binary"), or check out `8d0388b10a` and run `cargo build --release --bin curve25519_family` with `rustc 1.97.0`; a rebuild is not guaranteed to reproduce the frozen hash. Then run from this directory, in order:

1. `python3 scripts/run_grid.py`
2. `python3 scripts/census.py`
3. `python3 scripts/analyse.py`
4. `python3 scripts/tables.py`
5. `python3 scripts/figure.py`

Validation

An independent validator (a separate sub-agent that did not produce the study) checked commit `8c544a1829` and wrote `validator-report.json`, committed unchanged. Its status is "pass with errata": every measured number and preregistered verdict it checked recomputes, 56 of 70 cells replay identically with the frozen binary, and none of its 13 errata changes a preregistered verdict or a fitted number (as reported by the validator) [pcc:unverified]. The validation is an unsigned check, not a pcc claim. Each erratum was applied as follows.

id	issue	how it was applied
E1	"far above both thresholds" implied that m = 3 data exclude a lever	Answer, section 3 table (new column) and section 5 now give the d >= 9 fits 5.56 [−8.75, 19.88] and 4.51 [−1.27, 10.29] and the m = 4 fit 9.86 [−59.9, 79.6], and say that only the sensitivity fits have lower bounds above 0

E2	headline lacked the resource-limit caveat	Answer: "no lever at m = 2"; m = 3 and m = 4 undetermined because of 3.5 GB virtual memory, 1,200 s wall per target, shared host
E3	unit-test description	section 1: p = 211, the "consistent iff S_{m+1} has a root in V^m" property, `roots_of_unity_and_factors` listed, no m = 4 consistency test, log not committed
E4	L10 comparison mixed bases, ranges and units	section 3: L10 used the isogeny base; the G = 0 comparison ran the bit-interval base; slopes 5.5, 5.0 (tower mul-adds) and 5.15, 4.10 (field_ops) over |V| = 8 to 32; L10's guessing slope of about 1.5 is below this base's 2.77
E5	field_ops compared with rho group operations	section 8: ratios relabelled "field_ops per rho group operation"; no conversion factor recorded
E6	"solving degree d + 1 for prime d"	section 3: restricted to sym, m = 2 and 3; the 7 exceptions listed
E7	per-target symmetrisation unpriced	section 8 "Unpriced work": about 4.6e2, 3.1e4 and 6.2e7 multiplications at m = 2, 3, 4, exceeding T at m = 4 for d = 3 and 7 and at m = 3 for d = 3
E8	balanced totals from rounded s	section 4: 8.82, 10.33, 4.70 and 10.64 from the unrounded slopes
E9	"prediction m = 4 held"	section 5: undetermined for s; only the censoring expectation held
E10	post-data `tables.py` change undisclosed; k = 22 boundary cells	"Post-data change to a non-frozen script" paragraph; Answer names d = 3 and 5 for k = 22
E11	no host or load record; wall-clock censoring; time gaps	"Host, load and wall time" paragraph; wall time stated as not a claimed quantity
E12	binary only in a shared build directory; source and toolchain unpinned	"Source, toolchain and binary": commit `8d0388b10a`, `rustc 1.97.0 (2d8144b78 2026-07-07)` (read on the host after the run), binary copied to `/home/user/bin-archive/curve25519_family-dc2c30a4` with the frozen sha256; not committed; artifact-store upload still open
E13	untagged result sentences	sections 2 and 3: census, consistency, solving-degree and table-caption sentences tagged

The validator also reported the chain-length regression (section 3, exploratory) and the protocol-eligibility ambiguity (section 5); both are now in the text. Two items stay open: the binary is not yet in the artifact store, and the unit-test output is not committed (the tests were not rerun, because building was not allowed).